

In Utero Per- and Polyfluoroalkyl Substances (PFAS) Exposure and Changes in Infant T Helper Cell Development among UPSIDE-ECHO Cohort Participants

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ABSTRACT: **BACKGROUND:** Environmental exposures to toxicants, such as per- and polyfluoroalkyl substances (PFAS), during gestation can disrupt immune development, causing long-term impacts on a child's ability to generate a well-regulated, protective immune response. T-cells coordinate with all immune cell types to orchestrate both cellular and antibody-mediated responses. While there is compelling evidence that PFAS alters immunity in humans, the specific effects of early life PFAS exposure on infant T-cell development are unreported. Because of their central role in immunity, altered T-cell development in infants would have implications on immune responses broadly and long-term. **OBJECTIVES:** We seek to model longitudinal changes in the frequency of functionally distinct CD4+ T-cell subpopulations from birth through 12 months and their association with in utero PFAS exposure. **METHODS:** Maternal-infant dyads were recruited as part of the UPSIDE-ECHO cohort during the first trimester between 2015 and 2019 in Rochester, New York; dyads were followed through the infant's first birthday. Maternal PFAS concentrations (PFOS, PFOA, PFNA, PFHXS and PFDA) were quantified in serum during the second trimester using high-performance liquid chromatography and tandem mass spectrometry. Infant lymphocyte frequencies were assessed at birth, 6- and 12-months using mass cytometry and high-dimensional clustering methods. Linear mixed-effects models were employed to analyze the relationship between maternal PFAS concentrations and CD4+ T-cell subpopulations ($n = 200$). All models included a PFAS and age interaction and were adjusted for parity, infant sex, and prepregnancy body mass index. **RESULTS:** In utero PFAS exposure correlated with multiple CD4+ T-cell subpopulations in infants. The greatest effect sizes were seen in T follicular helper (Tfh) and T-helper 2 (Th2) cells at 12 months. A log₂-unit increase in PFOS was associated with lower Tfh [0.17% (95% CI: -0.30, -0.40)] and greater Th2 [0.27% (95% CI: 0.18, 0.35)] cell percentages at 12 months. Similar trends were observed for PFOA, PFNA, PFHXS, and PFDA. **DISCUSSION:** Maternal PFAS exposures correlate with cell-specific changes in the infant T-cell compartment, including key CD4+ T-cell subpopulations that play central roles in coordinating well-regulated protective immunity. Future studies into the role of PFAS-associated T-cell distribution and the risk of adverse immune-related health outcomes in children are warranted.

INTRODUCTION

The immune system development begins in the fetus during gestation and continues after birth. Different immune cell types arise from precursors, which can differentiate and develop in a regulated manner.^{1–3} Environmental exposure to harmful agents such as chemical toxicants or deficits in nutrition during gestation and infancy can dysregulate multiple developmental processes influencing health over a lifetime.^{4–7} Among these harmful agents, exposure to per- and polyfluoroalkyl substances (PFAS) stands out due to their widespread prevalence and persistence in the environment⁸ and their well-documented associations with several adverse health outcomes across the lifespan, including impaired reproduction, growth, neurodevelopment, and liver, kidney and immune dysfunction.^{9–14} PFAS are a group of synthetic chemicals characterized by interconnected carbon and fluorine (C–F) atom chains.⁸ The strong C–F bond makes PFAS resistant to environmental and physiological breakdown processes and allows them to

accumulate and persist in the environment and human tissues.¹⁵ In the U.S., over 97% of the population has detectable concentrations of PFAS in their blood,¹⁶ and similar findings have been reported in studies from Spain,¹⁷ Germany,¹⁸ and China.¹⁹ In addition to blood, PFAS have been detected in human lungs, brain, bone, liver, and kidneys.¹⁵ PFAS have also been detected in the placenta, cord blood, and breast milk,^{20–25} which further emphasizes the potential for early life exposures. Transplacental transfer^{20,21} and excretion through breastfeeding^{22,23,26} are main routes of early life PFAS exposure, and this transfer occurs during a critical window of development.

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One common observation is an inverse association between PFAS concentrations and antibody responses to vaccinations.^{9,10,27–31} While this inverse association between PFAS and antibodies has been consistent in children, the evidence for adults is less consistent across studies, PFAS species and vaccine types.^{32–37} It is unclear which immune system components are affected by PFAS to generate a net effect of lower humoral responses. The production of antibodies results from the collaboration between multiple types of immune cells.³⁸ When the immune system is stimulated by vaccinations or infections, cells from the innate compartment, such as macrophages and neutrophils, trigger inflammatory processes that activate cells of the adaptive immune system (T and B-cells), which can then establish immune memory. Immune memory is the ability of cells to retain antigen-specific information from prior exposures to facilitate a faster and more targeted response upon subsequent encounters. Antibody production by B-cells is influenced by CD4+ T-cells, which consist of different specialized subpopulations tailored to the immune response needed for a specific pathogen or antigen.³⁹ These include Th1, Th2, Th17, T follicular helper cells (Tfh), and T regulatory cells (Treg).⁴⁰ Each CD4+ T-cell subset carries a unique set of functions that influence B-cells by modulating the strength, specificity, and durability of their produced antibodies.⁴¹ For example, long-lived B-cells rely on Tfh cells for durable antibody production because Tfh cells promote germinal center formation in lymph nodes via IL-21 secretion.^{42–47} This interaction leads to B-cell proliferation, somatic hypermutation, and isotype class switching, ultimately resulting in high-affinity antibodies.^{42,44} Other CD4+ T-cell subpopulations modulate antibody class switching by fine-tuning the context of the immune response required.^{48–51} How early life exposures to PFAS affects these cell types is crucial for discovering how PFAS ultimately leads to changes in immune responses. This longitudinal study aims to identify alterations in CD4+ T-cell subpopulations related to developmental PFAS exposure as a first step in understanding pathways implicated in PFAS-related immune dysfunction.

METHODS

Study Design

Between December 2015 and April 2019, first-trimester pregnant women were recruited from outpatient clinics affiliated with the University of Rochester and enrolled in the Understanding Pregnancy Signals and Infant Development and Environmental Influences on Child Health Outcomes (UPSIDE-ECHO) study after informed consent. UPSIDE-ECHO is an ongoing prospective birth cohort designed to study prenatal exposures in relation to perinatal and child health outcomes in the Rochester, NY, area⁵² that is part of the National Institutes of Health (NIH) ECHO program.⁵³ Eligibility criteria included maternal age 18 years or older, singleton pregnancy, no known substance abuse problems or history of psychotic illness, and the ability to communicate in English. Participants with major endocrine disorders (e.g., polycystic ovary syndrome) and high-risk pregnancies at baseline were not enrolled; additionally, infants born prior to 37 weeks of gestation or with congenital anomalies or significant health problems were excluded ($n = 9$). Study visits, including physical exams, biospecimen collection, and questionnaire completion, were completed at each trimester and at birth, 6 and 12 months postnatally. This study protocol was approved by the University of Rochester School of Medicine and Dentistry Institutional Review Board (no. 58456).

Maternal and Infant Blood Collection

During the second trimester of pregnancy ($21 \pm \text{SD:1.8}$) gestational weeks, maternal serum from blood collected into EDTA tubes was frozen at -80°C until the time of analysis. For infant blood samples,

cord blood was collected by venipuncture into sodium heparin tubes immediately after birth. Infant blood was collected into sodium heparin tubes at 6 and 12 months. Peripheral blood mononuclear cells (PBMC) were isolated using a Ficoll-Paque density gradient (GE Healthcare – cytiva) and immediately cryopreserved in freezing medium consisting of 10% heat-inactivated Fetal Bovine Serum (FBS; Seradigm) and 10% DMSO (Amresco) at a cell concentration ranging from one to five million cells per mL.

PFAS Analysis

Maternal serum isolated from the second trimester of pregnancy was sent to Wadsworth Center's Human Health Exposure Analysis Resource (HHEAR) Laboratory Hub for PFAS measurement. PFAS were measured using high-performance liquid chromatography (HPLC) and tandem mass spectrometry (MS/MS) detection and quantification, as described in Honda et al. (2018).⁵⁴ Based on the PFAS measured by the Center for Disease Control and Prevention (CDC) National Health and Nutrition Examination Survey (NHANES) program, 14 PFAS commonly found in human blood samples were measured: NETFOSAA, NMFOSAA, PFBS, PFDA, PFDODA, PFHPA, PFHXA, PFHXS, PFNA, PFOA, PFOS, PFOSA, PFPEA, PFUNDA. The limit of detection (LOD) for individual PFAS varied from 0.02 to 0.03 ng/mL (Supplemental Table 1). To reduce exposure misclassification, PFAS with 80% detection rate or more, namely PFOA, PFOS, PFNA, PFHXS, and PFDA were included in the present analysis. For the PFAS that were detected in <100% of the participants, instrumentally derived values under the LOD were used as reported and values of 0 were replaced by the LOD divided by the square root of 2. Several quality control samples namely certified standard reference materials (NIST SRM 1957 and SRM 1958), HHEAR QC pools, procedural blanks and matrix spikes were included in the analysis. The coefficient of variation in concentrations of four PFAS in SRMs, QC pools and matrix spikes between batches was <10%.

Infant T-Cell Phenotyping

To control for batch effect and variability, all time points (birth, 6, and 12 months) from individual infant subjects were phenotyped in experimental batches that included 20 individual samples maximum. Each experimental batch also included one to two healthy adult PBMC controls that were carried across multiple batches. Thawing, stimulation, and staining of PBMCs for mass cytometry were performed as detailed previously Scheible et al. (2012) and McDavid et al. (2022).^{55,56} In brief, cryopreserved^{57–59} sample vials were thawed at 37°C in a water bath and washed twice with 10 mL of RPMI 1640 (Corning) supplemented with L-glutamine, 10% heat-inactivated Fetal Bovine Serum (FBS) (R10), and 25 U/uL Universal Nuclease (R10+n). Cord blood samples (CBMC) were treated with 3 mL 1x BD Pharm Lyse (BD Biosciences) supplemented with 25U/uL Universal Nuclease for 15 min to lyse red blood cells. Washed cells were rested overnight at 37°C in a 5% CO_2 incubator. After overnight rest, samples were counted on a dual-fluorescence cell counter (Cellometer K2; Nexcelom/Revvity); samples less than 70% viable were excluded from the assay (5.9% of samples at birth, 14.6% at 6 months and 12.7% at 12 months). If a sample was excluded due to viability and that participant had samples at other time points, the other-time point samples were included in the assay. Cells were resuspended in R10 at a concentration of 30 million-cells/mL, and 3 million cells were stimulated for 2 h with $2\ \mu\text{g/mL}$ of *Staphylococcus aureus*, Enterotoxin Type B (SEB) to activate T-cells. SEB triggers polyclonal T-cell activation and cytokine release, allowing the assessment of the T-cell functional properties in the absence of antigen-specific stimuli.⁶⁰ After the 2 h stimulation, the protein transport inhibitors GolgiPlug (BD Biosciences) and monensin (Sigma; 10 mM stock in EtOH) were added to allow for the accumulation of secreted proteins (i.e., cytokines) in the Golgi complex and enhance their detectability. After the addition of these reagents, the cells were incubated for 8 h at 37°C . Stimulated cells were washed with stain buffer (PBS + 2%FBS + 25 U/ μL Universal Nuclease) (SBn) and stained for 30 min at room temperature (RT) with unique combinations of metal-conjugated antihuman CD45 antibodies (Supplemental Table 2: Parameter Type, Barcodes) in a '6-choose-3'

or '7-choose-3' live-cell barcoding scheme.^{61–63} Barcoded samples were washed, resuspended in SBn, and then transferred into a single 15 mL culture tube. The pooled sample was filtered, centrifuged, and stained for 30 min at RT with a cocktail of metal-conjugated antihuman antibodies for surface molecules (Supplemental Table 2: Parameter Type, Surface). Next, cells were washed with SBn then stained with Cell-ID Cisplatin-194Pt (Fluidigm/Standard BioTools) at 500 nM/mL concentration for 5 min at RT. Cells were washed with SBn and fixed/permeabilized for 30 min at RT with Foxp3 Fixation-Permeabilization buffer (eBioscience) followed by a wash with Permeabilization Buffer (Perm-B) (eBioscience). Cells were intracellularly stained for 30 min at RT with Perm-B containing a cocktail of metal-conjugated antihuman antibodies (Supplemental Table 2: Parameter Type, Intracellular). Cells were washed, then fixed with 1 mL of 1.6% formaldehyde (Pierce) in PBS containing 500 nM of Cell-ID Intercalator-103Rh (Fluidigm/Standard BioTools) and incubated overnight at 4 °C. Fixed cells were washed with PBS, counted, and resuspended at a concentration of 13.75 million-cells/mL in filtered FBS with 10% DMSO (Amresco). Cells were aliquoted in Protein Lo-Bind tubes (Eppendorf) (2.75 million cells in 200 µL) and stored at –80 °C. Following aliquoting and freezing, individual aliquots were prepared for acquisition on a CyTOF 2 or Helios mass cytometer (Fluidigm/Standard BioTools) by washing twice with 2 mL ultrapure water and resuspending in 3 mL of ultrapure water containing 0.1x EQ Four Element Calibration Beads (Fluidigm/Standard BioTools). Aliquots were acquired using a Super Sampler (Victorian Airships).

Mass Cytometry Clustering Pipeline

After the acquisition, source data were normalized using the Normalization Passport in CyTOF Software with default settings (Fluidigm, 7.0.5189). Normalized data were analyzed using a custom R-based workflow (github.com/nlaniewski/SOMnambulate). Briefly, unique live-cell barcoded pools were debarcoded to generate sample-specific.fcs files. FlowSOM⁶⁴ was used to generate clusters representing major PBMC immune cell populations (Supporting Information Table 2: FlowSOM dimension, PBMC). Based on per-cluster feature expression, annotations were defined. Selecting for only CD3+, CD4+, and CD8+ clusters representing T-cells, a second FlowSOM was generated using an expanded set of both surface and intracellular markers to define T-cell-specific subclustering results (Supporting Information Table 2: FlowSOM dimension, T-cells). As with the PBMC clusters, percluster feature expressions were examined to define functional annotations. To aid in the interpretation of FlowSOM clusters, the nodes that comprise the self-organizing map (SOM) were visualized using uniform manifold approximation and projection (UMAP).⁶⁵ Default parameters were used to embed the nodes into two dimensions, using the same markers as those used to generate the respective SOM. Nodes (circles) were individually sized to represent the mean proportion of assigned cells and colored by their assigned-cluster annotation. For each FlowSOM result (PBMCs; T-cells), all cells were mapped on a per-sample basis, and the proportion of cells attributed to each cluster was calculated. The proportions used in subsequent visualizations and statistical models represent the percent of CD4+ T cells.

Measurement of Covariates and Selection of Potential Confounding Variables

After study enrollment in the first trimester, trained staff administered a questionnaire to obtain information on lifestyle, living environment, past pregnancies, medical conditions, medications used before, during, and after pregnancy, and sociodemographic information. Sociodemographic information, including self-reported maternal race and ethnicity, was collected at baseline, recognizing that these social constructs are associated with environmental exposures and maternal and child health outcomes.^{66,67} Infant birth weight and sex at birth were obtained from electronic health records. Confounding variables were selected using a directed acyclic graph (DAG) informed by previous reports demonstrating an association with PFAS and T-cells (Supplemental Figure 1).^{10,68} These were the categorical variables parity (0, 1 or 2+), infant sex (female or male), mode of delivery

(vaginal or cesarean section), maternal race (White, Black, Hispanic, Asian, Other), maternal education (high school or less, some college, bachelor's, postgraduate degree), and Medicaid during pregnancy (yes/no), and the continuous variables maternal age (years), prepregnancy body mass index (BMI) (kg/m²), annual income (dollars), and infant birth weight (grams). Prepregnancy BMI was calculated as $\frac{\text{pregnancy wt (lbs)}}{\text{height (in)}^2} \times 703$. The minimally sufficient set of variables from the DAG were parity, infant sex, and prepregnancy BMI.

Statistical Analysis

Descriptive characteristics of the entire sample are presented in Table 1. Specifically, frequencies and percentages for categorical variables, as well as means with standard deviations for continuous variables, are reported in this table. The Wilcoxon rank-sum test was used for comparing mean differences between two groups for a continuous variable, and the Wilcoxon signed-rank test was used to assess mean differences between two time points. The association between two continuous variables was evaluated using the Spearman's rank correlation test.

Owing to the longitudinal assessment of T-cell outcomes at multiple visits, two separate but related Linear Mixed-Effect regression (LMER) modeling strategies were employed to associate T-cell outcomes with individual PFAS concentrations, infant age, and their interactions. In both models, infant age is considered as a categorical variable with three levels (birth, 6 and 12 months), of which birth is considered as the baseline. In the first set of LMER models, a log-transformation designed to reduce the skewness is applied to the PFAS concentration levels: $x_{ij} = \log_2(\text{Conc}_{ij} + 1)$, where Conc_{ij} represents the original PFAS concentration level for the i th subject at the j th visit. In the second set of LMER models, PFAS concentration levels were dichotomized as "above median" or "below median", permitting the estimation of differences in T-cell outcomes over time as discrete categories according to the categorical interactions between infant age and binary PFAS concentration levels. We call the first set of models 'age-specific' models and the second set of models as 'differences-overtime' models.

For both approaches, a model is considered an *unadjusted model* if it includes only a PFAS level, infant age (categorical), and their interactions. It is considered an *adjusted model*, if it also includes the following additional fixed effects: parity (categorical), infant sex (categorical), and prepregnancy BMI (continuous). Additionally, a supplementary set of models that include the fixed effects in the *adjusted model* plus breastfeeding duration in days (continuous) were fitted, we call these *adjusted + breastfeeding duration (BFD)* models. A subject-specific random intercept was included in all models to account for serial correlation in the repeated T-cell measures. Unknown parameters in these models were estimated by the restricted maximum likelihood (REML) criterion. Regression t test implemented in R package *lmerTest*⁶⁹ was used to assess the statistical significance of a fixed effect of interest; F -tests as implemented in function *linearHypothesis()* in R package *car*⁷⁰ were used to test specific linear contrasts between infant ages.

In the age-specific models, marginal effects for the infant age and PFAS interaction were estimated using the R package *margins*,⁷¹ and pairwise comparisons between these predictions were assessed by the function *test_predictions()* from the R package *ggeffects*.⁷² In the difference-overtime models, the R package *lsmean*⁷³ was used to estimate the marginal means by infant age and categorical PFAS. Results for the difference-overtime models were presented as differences in marginal mean percent with corresponding 95% confidence intervals, and pairwise comparisons were performed using the function *contrast()* to contrast mean percentages between below and above median PFAS. Model assumptions, including normality, linearity, and homoscedasticity of the residuals, were visually evaluated by inspection of diagnostic plots, and assumptions were met. A p -value of less than 0.05 was considered statistically significant for individual hypothesis tests. For analyses involving multiple hypotheses, the Benjamini–Hochberg procedure was employed to control the false discovery rate (FDR) at a level of 0.05. All statistical analyses were conducted using R Foundation for Statistical Computing version 4.3.3.⁷⁴

Table 1. UPSIDE-ECHO Infant and Maternal Characteristics (N = 200)^a

characteristic	mean ± SD
Gestational age (weeks)	39.74 ± 1.11
Birth weight (grams)	3411.39 ± 464.49
Breastfeeding duration (days) (N = 183)	212.77 ± 162.99
BFD not reported (N = 17)	
Maternal age	N (%)
18–25	41 (20.5)
26–35	142 (71)
36+	17 (8.5)
Prepregnancy BMI	
Underweight	1 (0.5)
Normal Weight	90 (45)
Overweight	50 (25)
Obese	59 (29.5)
Annual income	
Lower class	41 (20.5)
Lower-Middle class	48 (24)
Middle class	40 (20)
Upper-Middle class	34 (17)
Upper class	19 (9.5)
Not reported	18 (9)
Parity	
0	69 (34.5)
1	78 (39)
2+	53 (26.5)
Infant sex	
Male	105 (52.5)
Female	95 (47.5)
Mode of delivery	
Vaginal	153 (76.5)
Caesarian	47 (23.5)
Maternal race	
White	125 (62.5)
Black	40 (20)
Hispanic	16 (8)
Asian	10 (5)
Other	9 (4.5)
Maternal education	
HS or less	64 (32)
Some college	31 (15.5)
Bachelors	54 (27)
Postgraduate	50 (25)
Not reported	1 (0.5)
Medicaid during pregnancy	
No	114 (57)
Yes	86 (43)

^aBMI: body mass index, SD: standard deviation, HS: high school, BFD: breastfeeding duration.

RESULTS

Study Population

The analysis included 200 full-term infants (Supplemental Figure 2). Participants were selected from the larger UPSIDE-ECHO cohort if they had second-trimester maternal PFAS measured and infant T-cell phenotyping from at least one time point (birth, 6, or 12 months of age). Cohort characteristics are shown in Table 1. In summary, 71% of participants were between 26 and 35 years old at enrollment; 62.5% of the cohort self-identified as White, 20% as Black, 8% as Hispanic, 5% as

Asian, and 4.5% as other. 52% reported having at least a bachelor's degree, and 43% were enrolled in Medicaid during pregnancy. Parity was categorized based on pregnancies preceding the current pregnancy (34.5% nulliparous, 39% primiparous, or 26.5% multiparous). The mean gestational age at birth was 39.74 ± SD:1.11 weeks, 76.5% of infants were born vaginally, and 52.5% were assigned male sex at birth. Among the 200 mother-infant pairs who contributed at least one observation to the current study, compared to birth, those with 6- or 12-month data tended to be older at the time of the child's birth, higher income and education, and more likely to report White race (Supplemental Table 3).

PFAS Exposure

PFAS concentrations measured in the participants included in this study (n = 200) are shown in Table 2. Five PFAS (PFOS,

Table 2. Distribution of Perfluoroalkyl Substances (PFAS) in Second-Trimester Maternal Plasma (ng/mL) in the UPSIDE-ECHO Study (N = 200)^a

PFAS	mean	SD	min	1st Q	median	3rd Q	max
PFOS	2.87	2.13	0.61	1.83	2.51	3.22	21.30
PFOA	0.68	0.51	0.06	0.40	0.59	0.83	5.45
PFNA	0.33	0.36	0.06	0.17	0.24	0.35	2.87
PFHXS	1.96	0.92	0.56	1.38	1.75	2.42	7.17
PFDA	0.08	0.11	<LOD	0.03	0.05	0.09	1.04

^aPFOS: perfluorooctanesulfonic acid, PFOA: perfluorooctanoic acid, PFHXS: perfluorohexanesulfonic acid, PFNA: perfluorononanoic acid, PFDA: perfluorodecanoic acid, SD: standard deviation, Min: minimum, Max: maximum, LOD: limit of detection, Q: quantile, LOD: limit of detection.

PFOA, PFNA, PFHXS, and PFDA) were detected in greater than 80% of the second-trimester maternal serum samples. The concentrations and distribution of the nine additional PFAS, which were detected in <80% of subjects, are in Supplemental Table 1. PFOS was present at the highest mean concentration (2.87 ± 2.13 ng/mL), followed by PFHXS (1.96 ± 0.92 ng/mL), PFOA (0.68 ± 0.51 ng/mL), and the lowest concentration being PFDA (0.08 ± 0.11 ng/mL). Relative to the NHANES general population concentrations, PFOS, PFOA, PFNA, and PFDA mean concentrations were lower.⁷⁵ PFOA and PFNA were the most strongly correlated (Rho = 0.64, p-value = 0.0001), followed by PFOS and PFNA (Rho = 0.58, p-value = 0.0002) and PFOS and PFOA (Rho = 0.53, p-value = 0.0005). PFHXS showed a moderate correlation with PFOS (Rho = 0.38, p-value = 0.021) and PFOA (Rho = 0.30, p-value = 0.049); PFNA and PFHXS were the most weakly correlated (Rho = 0.23, p-value = 0.113).

Determinants of exposure for the UPSIDE-ECHO population can be found in Supporting Information Table 4. Overall, the concentrations of all PFAS were lower with higher maternal parity. Maternal PFAS concentrations in serum were lower when the fetal sex was female. Generally, mean PFAS concentrations were higher in white mothers and mothers with higher income and education, and mean PFAS concentrations were lower in older mothers and mothers with higher prepregnancy BMI. Participants with Medicaid during pregnancy had lower concentrations of PFAS.

Infant T-Cell Clustering and Phenotyping

A 36-marker mass cytometry panel (Supplemental Table 2) was used to deeply immunophenotype infant immune cell

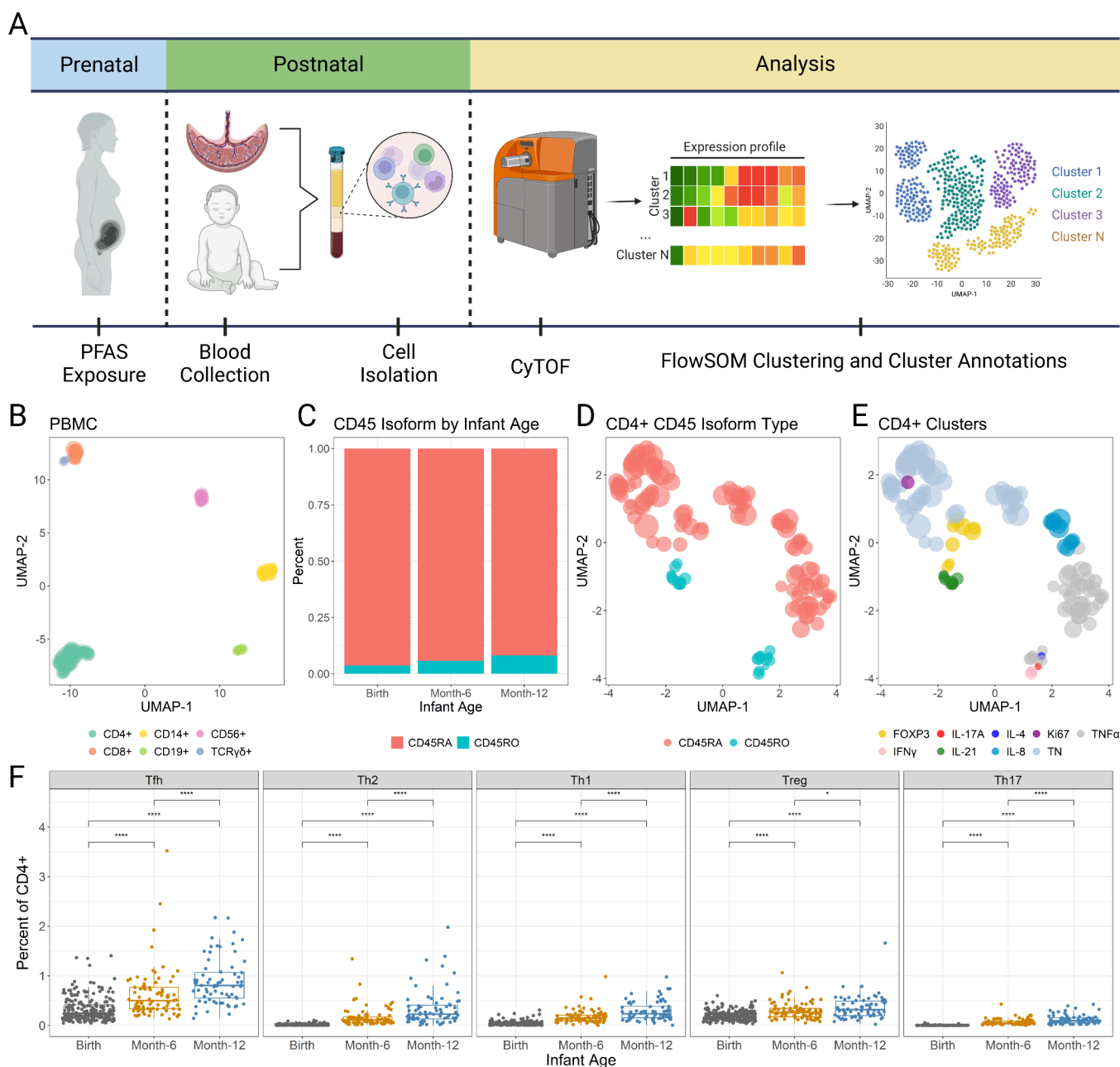


Figure 1. Identification of CD4⁺ T-cell subpopulations in UPSIDE-ECHO infants and their age-associated frequencies. (A) Study design timeline and T-cell analysis pipeline ($n = 200$). (B) Uniform Manifold Approximation and Projection (UMAP) representation of FlowSOM nodes colored by major peripheral blood mononuclear cells (PBMC): CD4⁺, CD8⁺, CD14⁺, CD19⁺, CD56⁺, TCRγδ⁺; (C) Stacked bar graph denoting the proportion of CD45RA⁺ and CD45RO⁺ CD4⁺ T-cell subpopulations at birth, 6 and 12 months of age. (D, E) UMAP representation of FlowSOM nodes colored by expression of (D) CD45 isoform type (CD45RA⁺ or CD45RO⁺) within CD4⁺ T-cell FlowSOM nodes, and (E) CD4⁺ T-cell subcluster annotations based on the following marker expression: Treg (FOXP3+CD127[−]), Th1 (IFNγ⁺), Th17 (IL-17A⁺), Tfh (IL-21⁺), Th2 (IL-4⁺), Naïve-cytokine-negative (TN). Circle size on UMAPs represents the relative mean count of total cells assigned to each FlowSOM node. (F) Frequency of CD4⁺ T-cell subpopulation in cord blood at birth and PBMC at six and 12 months. Each dot on the boxplots denotes data from an individual infant at the indicated age; scales for boxplots are fixed. Created in BioRender. Castro, D. (2025) <https://BioRender.com/Su3k0sw>.

populations over the first year of life. FlowSOM clustering algorithm was used to identify and group immune populations (Figure 1A). PBMC populations were identified as follows: $\gamma\delta$ T-cells were identified as CD3⁺ TCRγδ⁺ cells, cytotoxic T-cells as CD3⁺ CD8⁺, helper T-cells as CD3⁺ CD4⁺, B-cells as CD19⁺, NK cells as CD56⁺, and monocytes as CD14⁺ cells. UMAP visualization of infant leukocytes revealed separation between the major immune populations, including T-cells, B-cells, NK cells, and monocytes (Figure 1B).

To further phenotype CD4⁺ T cells and their age-associated changes, CD4⁺ T-cells from the first level of clustering (i.e., PBMC) were selected for a second level of FlowSOM clustering using an expanded set of phenotypic and functional markers to identify unique CD4⁺ T-cell subclusters. This subclustering analysis identified 29 unique CD4⁺ T-cell subclusters (Supplemental Figure 3A,B). During the first year of life, naïve (CD45RA⁺) comprised most of the CD4⁺ T-cell compartment, while T-cells carrying markers of immunological memory

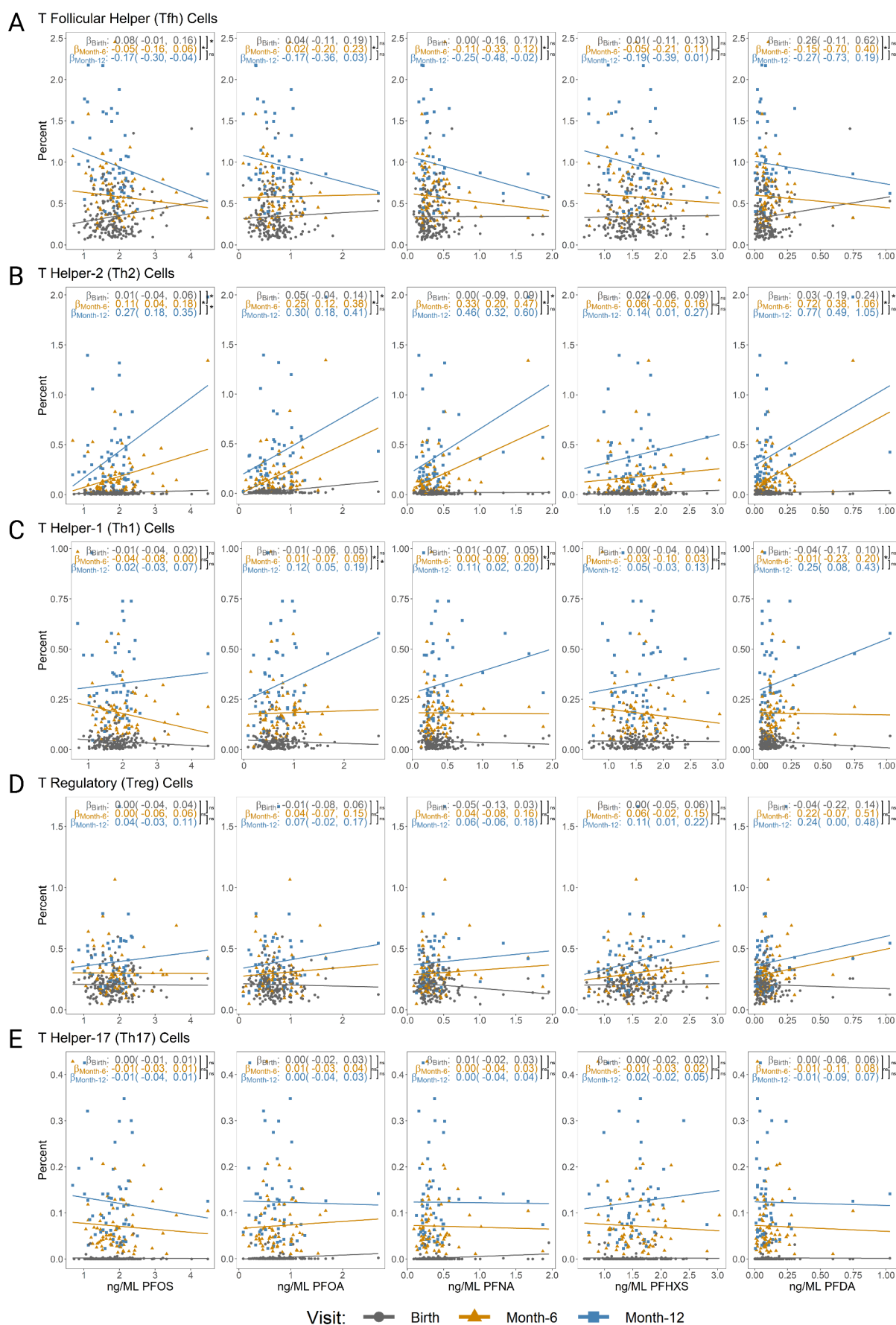


Figure 2. Second-trimester maternal PFAS concentrations associates with infant CD4⁺ T-cell subpopulation frequencies in an age-dependent manner in the UPSIDE-ECHO study. The plots illustrate the regression lines for the age-specific average marginal effects of the association between

Figure 2. continued

continuous $\log_2$ transformed maternal PFAS ($\log_2$ ng/mL) concentrations and infant CD4+ T-cell subpopulation percentages ($n = 200$). Rows represent individual CD4+ T-cells subpopulations: (A) T follicular helper cells, (B) T helper 2 cells, (C) T helper 1 cells, (D) T regulatory cells, and (E) T helper 17 cells. Columns represent individual PFAS; PFOS, PFOA, PFNA, PFHXS and PFDA. The colors and shapes represent the infant age: birth (gray circles), 6 months (orange triangles), and 12 months (blue squares). Each shape represents an individual infant at the indicated age. Age-specific average marginal effects (β) were adjusted for the categorical covariates parity (0, 1 and 2+), infant sex (female and male), and continuous prepregnancy body mass index (kg/m^2). Parentheses next to each age-specific β contains the 95% confidence interval for the estimate. Pairwise comparisons between the age-specific β s are represented by the brackets. Asterisks denote statistically significant (p -value < 0.05), and 'ns' indicates nonstatistically significant differences between age-specific β s. Axis are independent for each CD4+ T-cell subpopulation and each PFAS. Supplemental Tables 7, 8, and 9 contain corresponding numeric data.

(CD45RO+, CD45RA−) were a small fraction at birth and increased over time (Figure 1C,D). Within the memory (CD45RO+) compartment, canonical CD4+ T-cell subpopulations Th1 (IFN γ), Th2 (IL-4+), Th17 (IL-17A+), Tregs (FOXP3+ CD127−) and Tfh cells (IL-21+) were identified based on the expression of surface and intracellular markers that distinguish functional properties of these subpopulations (Figure 1E, Supplemental Figure 3B–C). All subsequent analyses were performed using the five CD4+ T-cell subclusters pertaining to the CD4+ T-cell subpopulations, as representatives of the canonical T-cell differentiation states (Tfh, Th2, Th1, Treg and Th17). The percentages of these five CD4+ T-cell subpopulations increased over time (Figure 1F, Supplemental Table 5). The distribution of the CD4+ T-cell subpopulation by infant age and cohort characteristics can be found in Supplemental Tables 5 and 6.

Maternal PFAS Exposure Associates with Infant CD4+ T-Cell Frequencies in an Age-Dependent Manner

Figure 2 shows the age-specific adjusted association for PFOS, PFOA, PFNA, PFHXS, and PFDA and each CD4+ T-cell subpopulation. The unadjusted and adjusted age-specific estimates for this model can be found in Supporting Information Table 7. Overall, the associations between PFAS and CD4+ T-cell subpopulations were stronger at 6- and 12-months of age versus birth; with statistical evidence of effect modification by infant age for Tfh, Th2, and Th1 cells, but not for Tregs or Th17 cells as denoted by the interaction p -values (Supporting Information Tables 8 and 9). These associations were consistent for the crude and adjusted models.

Tfh cells were inversely associated with maternal PFAS; each doubling of PFOS was associated with a 0.17 (95%CI of β : −0.30, −0.04) lower Tfh percentage measured at 12 months of age, a 0.05 (95%CI of β : −0.16, 0.06) lower Tfh percentage at 6 months, and a 0.08 (95%CI of β : −0.01, 0.16) greater Tfh percentage at birth (p -value for age interaction = 0.002). Similar trends were observed between Tfh percentages and PFOA, PFNA, PFHXS, and PFDA (Figure 2A).

In contrast, Th2, Th1, and Treg cells were positively associated with maternal PFAS exposure. For Th2 cells, each doubling of PFOA was associated with a 0.30 (95%CI of β : 0.18, 0.41) greater Th2 percentage measured at 12 months of age, a 0.25 (95%CI of β : 0.12, 0.38) greater Th2 percentage at 6 months, and a 0.05 (95%CI of β : −0.04, 0.14) greater Th2 percentage at birth (p -value for age interaction = 0.000). Similar trends were observed between Th2 percentages and PFOS, PFNA, PFHXS, and PFDA (Figure 2B). For Th1 cells, each doubling of PFNA was associated with a 0.11 (95%CI of β : 0.02, 0.20) greater Th1 percentage measured at 12 months of age and no association at 6 months of age or birth (p -value for age interaction = 0.054). Similar trends were observed between Th1 percentages and PFOS, PFOA, PFHXS, and PFDA (Figure 2C).

For Tregs, each doubling of PFHXS was associated with a 0.11 (95%CI of β : 0.01, 0.22) greater Treg percentage measured at 12 months of age, a 0.06 (95%CI of β : −0.02, 0.15) greater Treg percentage at 6 months, and no association at birth (p -value for age interaction = 0.129). Similar trends were observed between Treg percentages and PFOS, PFOA, PFNA, and PFDA (Figure 2D). No associations were observed between Th17 percentages and the doubling of PFOS, PFOA, PFNA, PFHXS, and PFDA (Figure 2E). The associations between each doubling of the PFAS evaluated (PFOS, PFOA, PFNA, PFHXS, and PFDA) and the cell percentages of Tfh, Th2, Th1, Tregs, and Th17 at birth, 6 and 12 months remained similar in magnitude and directionality after including breastfeeding duration in our main model (Supplementary Table 7).

Maternal PFAS Exposure Predicts CD4+ T-Cell Trajectories in Infants

All the T-cell subpopulation increased in frequency over time, regardless of their membership to either PFAS exposure group (high or low) in our differences-overtime models (Figure 3). Membership to the high exposure PFAS group was associated with higher frequencies of the Th2, Th1 and Treg subpopulations and lower frequencies of Tfh subpopulation. PFAS-dependent effects were more pronounced at later infant ages, especially 12 months of age. This was the case for Tfh, Th2, Th1, and Treg cells, but not for Th17 cells. Unadjusted and adjusted age-specific marginal means and pairwise comparisons for below and above median PFAS can be found in Supplemental Tables 10 and 11, respectively. For Tfh cells and PFOS exposure, the adjusted negative association was strongest at 12 months of age, with a mean value for above median PFOS of $0.86 \pm \text{SD}: 0.05$ (95%CI: 0.76, 0.96) versus a mean value for below median PFOS of $1.13 \pm \text{SD}: 0.06$ (95%CI: 1.02, 1.24) (p -value for difference = 0.002); similar trends were observed for PFOA and PFNA, and no differences were observed for PFHXS or PFDA (Figure 3A).

For Th2 cells and PFOA exposure, the adjusted positive association was present at 6 months but stronger at 12 months of age. At 12 months we observed a mean value for Th2 cells for above median PFOA of $0.44 \pm \text{SD}: 0.03$ (95%CI: 0.37, 0.50) versus a mean value for below median PFOA of $0.32 \pm \text{SD}: 0.04$ (95%CI: 0.25, 0.39) (p -value for difference = 0.002); similar trends were observed for PFOS, PFNA, PFHXS, and PFDA (Figure 3B). For Th1 cells and PFNA exposure, the adjusted positive association was observed at 12 months of age, with a mean value for above the median PFNA of $0.37 \pm \text{SD}: 0.02$ (95%CI: 0.32, 0.41) versus a mean value for below median PFNA of $0.30 \pm \text{SD}: 0.02$ (95%CI: 0.25, 0.34) (p -value for difference = 0.019); similar trends were observed for PFOS, PFOA, PFHXS, and PFDA (Figure 3C). For Treg cells and PFHXS exposure, the adjusted positive association was present at 6 months but stronger at 12 months of age; at 12 months, we observed a Treg

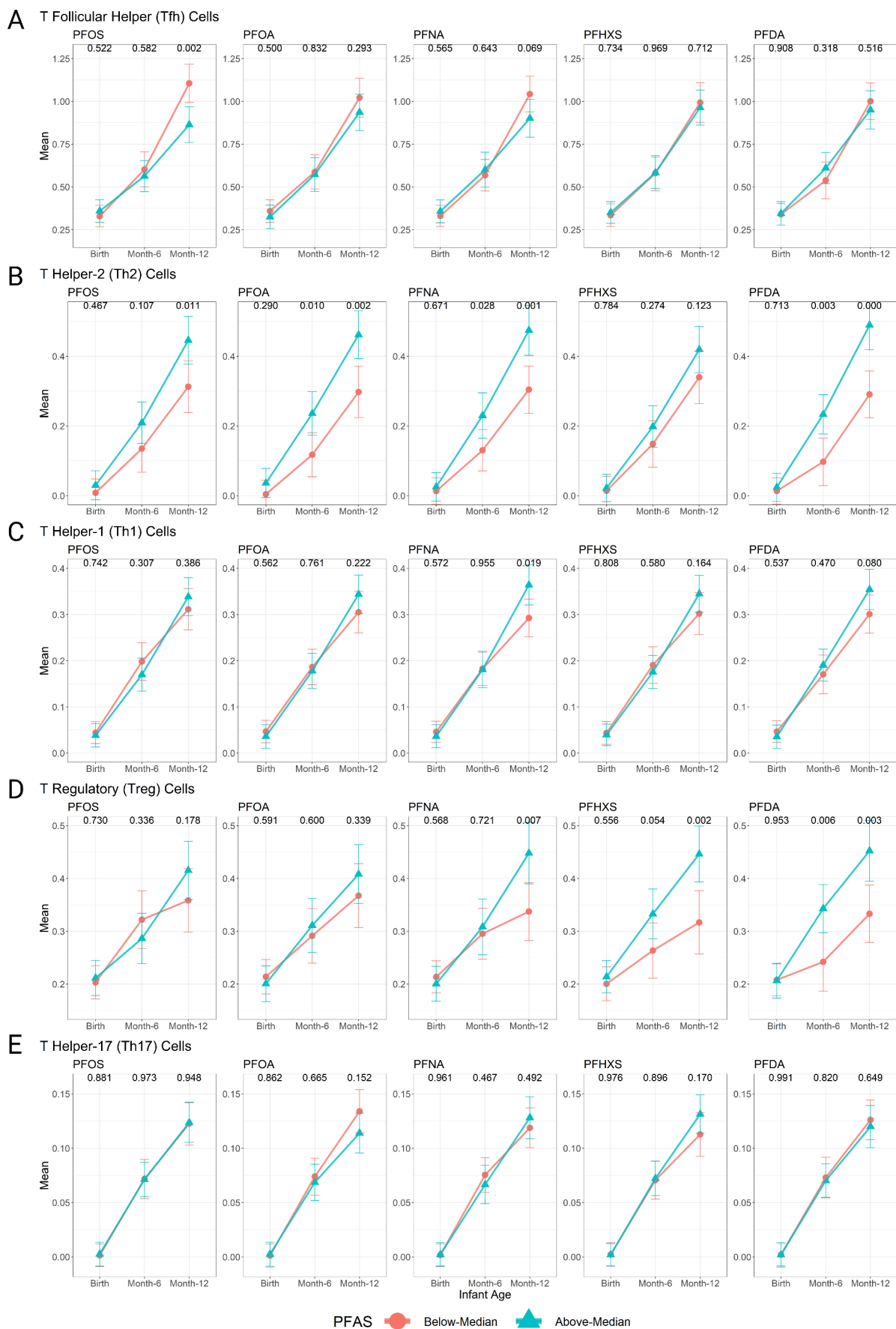


Figure 3. Second trimester maternal PFAS concentrations predicts CD4⁺ T-cell subpopulation trajectories in infants in the UPSIDE-ECHO study. The plots illustrate differences-overtime between infant CD4⁺ T-cell subpopulations by a categorical PFAS value (below versus above median value; *n*

Figure 3. continued

= 200). Rows represent individual CD4+ T-cell subpopulations: (A) T follicular helper cells, (B) T helper 2 cells, (C) T helper 1 cells, (D) T regulatory cells, and (E) T helper 17 cells. Columns represent individual PFAS: PFOS, PFOA, PFNA, PFHXS, and PFDA). The colors and shapes represent the categorical PFAS value: below the median (pink-circles) and above the median (blue-triangles). Shapes denote the predicted marginal mean of each CD4+ T-cell subpopulation at the indicated age. Marginal means were adjusted for the categorical covariates parity (0, 1, 2+), infant sex (female or male), and continuous prepregnancy body mass index (kg/m^2). Error bars represent the standard error of the marginal means. P-values denote the statistical significance between marginal means contrasting below vs above the median PFAS; the contrast was considered statistically significant if the p -value < 0.05. The y-axis is independent for each CD4+ T-cell subpopulation. Supplemental Tables 10, 11, and 12 contain corresponding numeric data. Number of participants by categorical PFAS value (A = above median, B = below median): PFOS: A = 101, B = 99; PFOA: A = 100, B = 100; PFNA: A = 95, B = 105; PFHXS: A = 106, B = 94; PFDA: A = 95, B = 105.

mean value for above the median PFHXS of $0.45 \pm \text{SD}: 0.03$ (95%CI: 0.40, 0.50) versus a mean value for below the median PFHXS of $0.32 \pm \text{SD}: 0.03$ (95%CI: 0.26, 0.38) (p -value for difference = 0.002); similar trends for 12 months but not 6 months were observed for PFOS, PFOA, PFNA, and PFDA (Figure 3D). No differences between the mean value of Th17 cells for above versus below median PFOS, PFOA, PFNA, PFHXS, and PFDA were observed (Figure 3E).

The mean percentages of Tfh, Th2, Th1, Treg, and Th17, after adjusting for breastfeeding duration, were similar to the mean percentages in the main differences-overtime adjusted model not adjusting for breastfeeding duration (Supporting Information Table 10). However, when contrasting between below vs above the median PFAS groups, there appears to be an enhanced statistical significance for the differences in the mean percentages of the T-cell subpopulations by PFAS group, mainly for the 12 month age.

DISCUSSION

In this prospective cohort study, we investigated the association between in utero PFAS exposure and CD4+ T-cell subpopulations in infants over the first year of life. The potential of PFASs to affect the immune system has been widely reported in population studies across the world, with consistently observed reductions in antibody responses to common childhood vaccinations.^{9,12} Here we showed that in utero PFAS exposure associates with specific CD4+ T-cell subpopulations during the first year of life; with increased frequencies of the Th2, Th1, and Tregs subpopulations and reduced frequencies of the Tfh subpopulation.

PFAS-associated reduction in vaccine responses are consistently reported for vaccines that induce T-cell dependent antibody responses,⁷⁶ including tetanus, diphtheria, measles, rubella, and mumps.^{27,29,77–79} This indicates that T-cells are a logical focus for research on PFAS immunomodulation. The evidence of PFAS-associated cellular immunomodulation is limited in both humans and animal studies. In humans, Lopez-Espinoza et al. in 2021⁸⁰ evaluated the frequency of total peripheral lymphocyte counts in adults as a function of PFAS exposure, finding increases in total lymphocytes and total CD4+ T-cells. Additionally, animal studies have shown PFAS-associated modulation of T-cell dependent antibody responses,^{81–83} changes in T-cell cytokines expression,^{84,85} and changes in the frequencies of total lymphocytes.^{86,87} While these findings are informative, without knowledge of the many potential specific CD4+ T-cell functions that could be affected by PFAS exposure, it is difficult to decipher mechanisms and predict the impact on early immune responses. The findings reported here address this, showing that exposure to PFAS in utero alters the proportion of functionally distinct T cell populations in infants.

Specifically, we found that increased concentrations of maternal PFAS were associated with a reduction in the percentage of Tfh cells, increased Th2, Th1, and Treg populations, and no difference in Th17 cells. Moreover, each association was stronger from birth to 12 months of age. These findings indicate that the potential effects of prenatal PFAS exposure may become more pronounced as infants grow older. This could be attributed to ongoing T-cell development during the first year of life and that certain T-cell subpopulations may not emerge until the infant is exposed to specific stimuli. This further suggests that as time passes and T-cell subpopulations continue to develop, the impact of PFAS exposure on the cells becomes more apparent.

The immune system protects against pathogens through a coordinated network of cells, tissues, and organs. CD4+ T-cells orchestrate the immune response, making it specific to the type of stimuli received and aiding in the generation and maintenance of immune memory. Particularly, Tfh cells help direct B-cells to produce long-lasting, high-affinity antibodies essential for immune memory.⁴² This study revealed a negative association between the percentage of Tfh cells and maternal PFAS exposure during pregnancy. A reduction in Tfh cells in infants is consistent with the observed reductions in antibodies to childhood vaccinations.^{28,29,77,78,88–90} Our findings provides potential mechanistic insights into this observation because Tfh cells are essential for mature and durable antibody responses.^{42,46,47,91}

Given the complexity of antibody responses, it is unlikely that Tfh cells are the only immune cell type altered by PFAS exposure that could result in a reduction of antibodies. For example, we also observed that Tregs were enhanced with increased maternal PFAS concentrations. Tregs are recognized for their immunosuppressive functions and their capacity to inhibit the activation and proliferation of effector T-cells.^{92,93} Studies have demonstrated that Tregs can reduce immune responses to vaccinations by suppressing effector responses.^{94,95} An increase in Treg cells could suppress T-cell or B-cell responses, thereby attenuating antibody production. More generally, an enrichment in Treg cells could support an immunosuppressed environment, dampening the overall ability to fight infections and generate sufficient antibody titers, as observed in association with PFAS exposure.

In addition to Tfh and Tregs, our study found a positive association between the percent of Th1 and Th2 cells with maternal PFAS concentrations. While Th1 and Th2 cells play important defense roles during infections, excessive Th1 and Th2 activity can be involved in the immunopathogenesis of several chronic health conditions, including allergy and autoimmunity.^{96–101} Th1 cells are among the predominant T-cell subpopulations in systemic autoimmune disorders, such as rheumatoid arthritis,¹⁰² as well as organ-specific autoimmunity,

such as Type 1 diabetes mellitus,¹⁰³ and other debilitating inflammatory disorders such as Crohn's disease.¹⁰⁴ On the other hand, Th2 cells have been associated with the development of atopic and allergic diseases such as atopic dermatitis, eczema, chronic rhinosinusitis, and asthma.^{101,105,106} While studies have associated PFAS exposure with allergic or autoimmune outcomes in children, the findings are inconsistent, and some show no significant association.^{107–118} Given such inconsistencies, it is likely that any biological relationship is nuanced and downstream of several potential specific pathways. For instance, several studies have examined the association between PFAS exposure and allergic outcomes in infants by measuring IgE levels in umbilical cord serum. For example, Okada et al. in 2012¹¹¹ reported lower umbilical cord IgE levels at birth associated with higher maternal PFOA, whereas Wang et al. in 2011¹¹² found increased IgE levels in cord blood associated with prenatal PFOA and PFOS exposure. IgE plays an important role in the allergic response; however, baseline total IgE levels may not reliably predict allergic outcomes, particularly because they are not specific to a particular allergen. Additionally, a study by Oulhote et al. 2017¹¹⁹ found a positive association between PFAS concentrations and basophil counts in 5-year-old infants; while their study did not investigate allergic outcomes, it is well established that basophils play a crucial role in initiating Th2-mediated immunity.¹²⁰ Thus, an enhancement in Th2 responses, as we observe in our results, operates in concert with these innate immune cells, which could lead to the development of adverse allergic outcomes. Further, type 2 cytokines such as IL-4, IL-5, and IL-13 have been associated with asthma development, and Th2 cells are the primary source of these cytokines.^{99,105,106} Similar to allergic outcomes, evidence for the association between PFAS exposure and asthma has been mixed. Studies reporting negative associations between exposure to PFAS, and the risk of asthma were mainly conducted in children under the age of 10,^{115,121} while those reporting a positive association were conducted in adolescents.^{116,118,122} These discrepancies emphasize the importance of considering age and longitudinal analysis when studying immune-related outcomes in children. Our study demonstrates that the effects of PFAS exposure may not be detectable or have an impact until the time during which a specific T-cell subpopulation emerges or is called into action postnatally.

Our findings reveal a 0.30 (95% CI: 0.18, 0.41) greater Th2 percentage associated with PFOA at 12 months. While the changes observed in the age-specific percentages of CD4+ T-cell subpopulations in this study may appear small, these variation sizes have had health implications in previous studies. For instance, an estimated 1.5–2-fold increase in Th2 cells has been noted among infant patients with atopic dermatitis or bronchial asthma compared to healthy controls.^{123,124} Using the mean Th2 percentage of 0.38% ($\pm$ SD: 0.38) as a reference in our infants (Figure 1F, Supplemental Table S), an increase of 0.30 indicates an overall 79% increase in Th2 cells. Further, numerous studies have shown that minor changes in the frequency of circulating T follicular helper (cTfh) cells correlate with antigen-specific antibody levels in humans,^{46,125–127} at ranges similar to the 18% reduction in Tfh cells we observed. Individuals with higher frequencies of cTfh cells typically exhibit more robust antibody responses; this relationship has been observed in vaccine responses to hepatitis B,¹²⁷ yellow fever virus,¹²⁶ and influenza,¹²⁵ as well as among children infected with malaria.⁴⁶ Our results provide strong evidence that maternal PFAS exposure affects the infant's immune system

and supports the idea that PFAS may modulate immune function in human infants.

This study has several strengths, as well as some limitations. One major strength is that this is a longitudinal study with deep immunophenotyping, high-dimensional analysis, and identification of infant CD4+ T-cell subpopulations. Additionally, since only generally healthy pregnancies and full-term healthy infants were enrolled in the study, we partly removed the potential for congenital-disease-related changes in CD4+ T-cell subpopulations that could influence immunological outcomes. Although our study sample was relatively small ($n = 200$) compared to larger cohort studies that showed PFAS-related antibody suppression,^{27,28} our study's in-depth immunophenotyping provides additional insight into one potential mechanism leading to antibody suppression.

It is well established that PFAS can cross the placenta^{20,128} and can be detected in the fetus;²¹ in the current study, however, PFAS were only measured during the second trimester of pregnancy and not directly in the infants. Measuring PFAS concentrations directly in infants would provide additional insights into postnatal PFAS exposure and its immunomodulatory influences on postnatal immune development. However, given that human T-cell development begins around the eighth week of pregnancy and continues after birth during infancy, exposures during the second trimester, as measured in our study, can plausibly impact the T-cell profile of the offspring.^{2,129,130} The potential immune consequences of prenatal pollutant exposure are well-documented; for example, human cohort studies have linked in utero exposure to persistent pollutants, such as polychlorinated biphenyls (PCB) and 2,3,7,8-tetrachlorodibenzodioxin (TCDD), with decreased antibody responses in infants and children.^{29,131,132} Although studies associating in utero exposure and lymphocytes in humans are limited, there are reports of early life exposures to dioxin and PCBs associated with changes in total lymphocyte counts;^{133,134} for example, Weisglas-Kuperus et al. 2000¹³⁴ estimated prenatal exposures to PCBs and dioxin and evaluated total lymphocyte counts in preschool children, finding an increase in total lymphocytes, including total CD4+ T-cells. Further evidence that developmental exposures shape T-cell function is provided by experimental studies, for example, Boule et al. 2014 using reciprocal adoptive transfer models found reductions in CD4+ T-cell subpopulations in adult mice after developmental exposure to TCDD, providing strong evidence that in utero exposure can significantly and durably alter T-cell function later in life.¹³⁵ To our knowledge, there are no studies evaluating the association between PFAS or other persistent pollutants and detailed immunophenotyping of the CD4+ T-cell subpopulations in humans as demonstrated in our study.

Our results indicate stronger associations between maternal PFAS exposure and CD4+ T-cell subpopulations later during infancy compared with birth. This is likely due to the fact that T-cells are still developing postnatally, and T-cell activation and differentiation are dependent on postnatal exposures. It is not unexpected, therefore, that PFAS-related variation in subpopulations would be measurable only after these CD4+ T-cell subpopulations emerge. Because of ongoing postnatal development of the T-cell compartment, PFAS exposure after birth is a key factor to be considered. Recently, PFAS have been detected in human breast milk^{22,23} and could represent a continuous source of maternally derived PFAS exposure in early postnatal life. We adjusted for breastfeeding duration and evaluated the association between maternal PFAS concentrations and CD4+

T-cell subpopulations. Our age-specific models were not significantly changed after adjusting for breastmilk exposure (Supporting Information Table 7). However, in our differences-over-time models, when contrasting between below vs above the median PFAS groups (Supporting Information Table 10), there appears to be an enhanced statistical significance for contrast at the 12 month time point. However, the current study was not designed to more directly test the independent effects of breastfeeding on the associations between CD4+ T cell subpopulations and PFAS exposure. While this potential enhancement in the magnitude of the effects after adjusting for breastfeeding exposure is of interest, these results should be considered carefully, as we do not have the statistical power to confidently assess the role of breastfeeding in the associations between PFAS and T-cell subpopulations. The supplementary models adjusting for breastmilk exposure were created with a subset of the participants that reported breastfeeding behaviors, reducing the statistical power of the models and increasing the risk of bias in our analysis. Rigorous studies to quantify the variation of PFAS in breast milk and formula or other postnatal exposures and immune development are warranted.

It is worth noting that participants in the study had lower mean serum concentrations of PFOS, PFOA, PFNA, and PFDA but higher PFHXS compared to the general population levels reported by NHANES (2017–2018).⁷⁵ Given the declining levels of legacy PFAS over time, our study provides insights for ongoing risk assessment of PFAS exposures⁷⁵ and suggests immunotoxic impacts in early life, even at these lower levels.

In conclusion, we report here an association between in utero exposure to PFAS and a decreased percent of Tfh cells and increased percentages of Th2, Th1, and Treg cells during the first year of life. These observations support the idea that there are immunomodulatory effects of developmental PFAS exposure. These findings contribute to our understanding of potential mechanisms by which PFAS exposure lowers antibody titers in infants and can inform future studies seeking to understand specific mechanisms of PFAS immunomodulation. Moreover, given that imbalances in T-cell subpopulations are associated with other diseases, in-depth profiling of infant T-cells suggests that in utero PFAS exposure may predispose to future immunopathogenesis. Future studies examining T-cell-mediated outcomes such as infection, autoimmunity, or inflammatory diseases in children with higher PFAS exposures are warranted.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/EHP.6c00037>.

Supplemental figures and tables (PDF)

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Notes

The authors declare no competing financial interest.

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